



Proposed CEA Link — December 2026

Clean Final Tuned with a tenth Sunday shift added and the wheel reordered — fatigue factors 3 → 2, longest run 9 → 6, and Sunday's fit down by a further fifth

Prepared 22 September 2026 · re-rendered 24 September 2026 · built for the December 2026 timetable and assessed by the workspace's own rule modules (Marylebone Roster v24.22) · figures on this page are computed, not typed

PROPOSAL

Clean Final Ten

A tenth Sunday shift and a reordered wheel - fewer factors, a shorter run

CODE TN-24-R7

FINGERPRINT 84b60df9

BUILT FROM duty table Clean Final Tuned + 1 cell · seed 7 (reorder) · 24 lines · 4 cover weeks

LINEAGE DERIVED: "Clean Final Tuned" (CFT-24-M3 · ae1a15bd) with ONE cell added — a tenth Sunday duty, 15:25–23:25 — and the wheel then reordered, whole weeks only. Every week pattern in the parent is still here unedited (the one exception being the week that gained the Sunday cell); what changed is which line number each week sits on. R7 is the reorder search's seed and it reproduces this grid exactly.

✓ 24 lines designed

✓ 35h 00m a week — exactly the contract

✓ 0 hard-limit breaches

✓ 0 rests under 12h

⚠ 2 fatigue factors present (today: 5)

11 of 24

shift times are today's

13 times are new; the rest are turns people already work

6 days

longest run of worked days

today's link reaches 9; Chiltern's limit is 13

2 of 20

working weeks are one turn

same clock time all week, as 7 of today's 16 are

5 in 24

full weekends off

21% of the rotation, against 20% today (4 in 20)

4 cover weeks

lines 1, 7, 12, 17

four weeks at lines 1, 7, 12 and 17 — gaps of 8, 6, 5, 5, exactly as the parent has them: the reorder holds the cover weeks fixed and swaps only working weeks

3h 10m a week

typical week-to-week move

how far the working day shifts each week — today's link moves 4h 0m a week

What this is

A proposal for the CEA link once the December 2026 timetable arrives, for a team of 24 rather than 20. The brief was to keep the **feel** of the link people work today — a week is one turn, rest days come in pairs, four whole cover weeks spread round the rotation, and the same 06:20 / 08:00 / 14:00 / 15:15 clock times — while meeting the staffing shape agreed for December 2026 (four to open every day, three through to the close, four on a Saturday, five still on at 22:00, fourteen on a Saturday, ten on a Sunday) and every rule the link is assessed against: Chiltern's 13-day limit, twelve hours between duties, the exact 35-hour contracted week, and the ORR's good-practice fatigue factors.

This is **"Clean Final Tuned" with one cell added** — a tenth Sunday duty, 15:25–23:25 — **and the wheel then reordered**, whole weeks only, so that the added Sunday does not lengthen anybody's run. Ten on a Sunday is the December 2026 headcount, met for the first time in this family.

Measured against the December 2026 timetable curve the way every proposal here is measured — the squared gap between each hour's share of demand and its share of cover, lower being better — **Sunday falls from 44.9 to 35.3**; Saturday stays at 20.5 and the five weekdays at Mon 28.4 · Tue 43.6 · Wed 42.3 · Thu 48.8 · Fri 31.0, because no weekday or Saturday cell was touched. **Fatigue factors present go from three to two**, the longest worked run from nine days to six, and full weekends off from four to five. The *Wk fit* column opposite is the fit of the average weekday, so it reads the same for this design, its parent and its grandparent — no weekday cell moved between them; page 5 carries each day's own figure.

What it is not. Not a recommendation and not a finished roster: nothing here has been through the roster office or a rep. The 24-line length, the four cover weeks and the December 2026 headcounts are owner-relayed figures with no document behind them (evidence class C), and the 13-day limit's policy citation is still outstanding — pages 6 and 7 say so where it applies. It clears every rule the tool can check, which is the floor for being worth discussing.

BEFORE THE FIGURES

The decision this sheet supports

What is being decided, which criteria are hard and which are the room's, how much each figure can bear, and what no page here can tell you. Read this before the tiles.

- 1 What this is
- 2 The decision it supports (this page)
- 3 The 24-line grid, with the words explained
- 4 A week compared with today
- 5 People on duty, hour by hour
- 6 Hard limits and the December 2026 figures
- 7 The ORR fatigue factors
- 8 Where it came from
- 9 The grid, ready to paste into the app

The question

Should the link for the December 2026 timetable be the hand-built Weekday Lates line, in this revision, accepting 12 on a Saturday against the agreed 14 and 1 of the four December 2026 headcount rules met, for a week shape its authors chose?

Clean Final Ten is read two ways: against **Quarter To (QT-24-Q34)** and **By the Book (BB-24-D7)**, the two searched answers; and within its own line, against the revision before it.

The criteria, in three tiers

HARD — A DESIGN MEETS THESE OR CANNOT BE RUN

- ✓ Twelve hours between duties — met, tightest 12h 30m
- ✓ Chiltern's 13-day limit — longest run 6 days
- ✓ The 35-hour contract, exactly — 35h 00m

SOFT — THE ROOM WEIGHS THESE AGAINST EACH OTHER

- December 2026 headcounts met: **1 of 4** (today 0 of 4)
- Fatigue factors present: **2** (today 5)
- Demand fit weekday · Sat · Sun: **34.4 · 20.5 · 35.3** (today 44.7 · 45.7 · 65.9; lower is closer)
- Times worked today: **11 of 24** · full weekends off: **5** in 24 (today 4 in 20)

PREFERENCE — STAFF'S TO STATE, NOT THE TOOL'S

- Early against late: earlies 7h 15m–8h 30m, lates 7h 30m–8h 55m
- 13 of 24 clock times are new — the duty table on page 4 lists them
- The longest duty, 8h 55m, and who holds it
- Cover weeks at lines 1, 7, 12, 17

How much to trust each figure

FIGURE ON THESE PAGES	WHERE IT COMES FROM	WEIGHT
0 rests under 12h · 35h 00m contract · 12/10 at the weekend	Computed from the cells; re-computed by the workspace on import	Firm
Longest run 6 days, inside 13	Computed; the 13-day limit's policy citation is still outstanding	Firm figure , unconfirmed threshold
Four at the open, three to close, 14 and 10 at the weekend	Owner-relayed December 2026 headcounts with no document behind them (class C)	Provisional
2 fatigue factors present	The ORR's 2021 list, which it states are not limits; 4 factors still "definition to confirm"	Advisory only — never pass or fail
Demand fit 34.4 · 20.5 · 35.3	Measured timetable, one share-based measure on one December 2026 timetable curve	Indicative
11 of 24 times are today's	Computed against the live 20-line link	Firm , but a proxy for acceptability

What no page here can tell you

- Whether the people who would work them accept a table in which 13 of 24 times are new (page 4 lists them) — familiarity is measured as "times worked today", which is a proxy for acceptability, not acceptability.
- Whether the roster clerk splits a cover week day-on-day-off. It is the one thing that moves the 48-hour-break figure (12 as a block, 12 split), and the link does not decide it.
- How leave and sickness cover land on a 24-line link with four cover weeks (lines 1, 7, 12, 17); nothing here models absence.
- Whether the December 2026 timetable demand curve holds on Wembley event days — the Saturday weighting approximates them, and the curve is a measured timetable, not a footfall count.

Before it is frozen

Lines 2, 3, 4, 5, 6 hold the times nobody works today; the people who would work those weeks read them first and say in one line each what they would change. The roster office answers the cover-week question. Someone confirms the source of the 13-day limit. Only then does the factor count on page 7 mean anything, and only then is this sheet more than an aid to a conversation.

THE ROTATION

The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

LINE	SUN	MON	TUE	WED	THU	FRI	SAT	MON-SAT hours	ALL WEEK hours	DAYS WORKED Mon-Sat	TURNS Mon-Fri
1	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
2	07:15 15:15	06:20 13:45	RD	RD	13:30 22:00	16:25 23:55	13:30 21:30	31h 25m	39h 25m	4	3
3	14:55 23:25	13:30 22:00	RD	13:30 22:00	13:30 22:00	13:00 21:00	14:30 22:30	41h 30m	50h 00m	5	2
4	RD	16:25 23:55	16:25 23:55	RD	RD	07:00 15:30	06:20 14:30	31h 40m	31h 40m	4	2
5	15:25 23:25	RD	06:20 13:45	06:20 13:45	07:00 15:30	RD	06:20 14:15	31h 15m	39h 15m	4	2
6	14:30 23:25	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	RD	RD	30h 00m	38h 55m	4	1
7	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
8	14:30 23:25	RD	06:20 13:45	06:20 13:45	06:20 13:45	06:20 13:45	RD	29h 40m	38h 35m	4	1
9	RD	12:00 20:30	12:00 20:30	12:00 20:30	12:00 20:30	12:00 20:30	RD	42h 30m	42h 30m	5	1
10	RD	08:30 17:00	RD	16:25 23:55	16:25 23:55	16:25 23:55	15:15 23:55	39h 40m	39h 40m	5	2
11	RD	14:00 22:30	13:30 22:00	13:30 22:00	13:30 22:00	RD	RD	34h 00m	34h 00m	4	2
12	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
13	07:15 15:45	06:20 13:45	RD	RD	13:30 22:00	14:00 22:30	11:00 18:30	31h 55m	40h 25m	4	3
14	10:30 19:00	16:25 23:55	RD	RD	06:20 14:20	06:20 14:20	06:20 14:30	31h 40m	40h 10m	4	2
15	07:15 15:45	RD	06:20 13:35	06:20 13:35	06:20 13:35	07:00 15:30	RD	30h 15m	38h 45m	4	2
16	RD	06:20 13:45	08:30 17:00	08:30 17:00	RD	RD	RD	24h 25m	24h 25m	3	2
17	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
18	07:15 15:45	RD	08:30 16:30	06:20 14:20	06:20 14:20	06:20 14:20	06:20 14:20	40h 00m	48h 30m	5	2
19	RD	RD	14:00 22:30	13:30 22:00	14:00 22:30	14:00 22:30	14:30 22:30	42h 00m	42h 00m	5	2
20	14:30 23:25	RD	14:00 22:30	14:00 22:30	14:00 22:30	16:25 23:55	RD	33h 00m	41h 55m	4	2
21	RD	14:00 22:30	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	15:15 23:55	47h 10m	47h 10m	6	2
22	RD	06:20 14:20	07:00 15:30	07:00 15:30	07:00 15:30	08:30 17:00	RD	42h 00m	42h 00m	5	3
23	RD	07:00 15:30	07:00 15:30	RD	RD	06:20 13:35	06:20 14:50	32h 45m	32h 45m	4	2
24	RD	RD	06:20 14:20	08:30 16:30	RD	13:30 22:00	15:15 23:55	33h 10m	33h 10m	4	3
Cover	10 SP 4	13 SP 4	15 SP 4	15 SP 4	16 SP 4	16 SP 4	12 SP 4	35h 00m	38h 33m	4.35	2 one turn
Minutes	5,115	6,225	7,205	7,205	7,780	7,690	5,895	42,000	47,115		

Early turns Late turns Cover (spare) week — four duties of seven, any turn Rest day

Mon-Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines · Turns: distinct clock times Monday to Friday, **bold** where the week is one turn (one clock time Mon-Fri and one early-or-late family across every worked day — the page 1 figure) · Minutes: duty minutes per day, and Mon-Sat is the contract (20 × 35h = 42,000)

The words on this page

Link — the whole rota: 24 weeks laid out as 24 lines. Everyone works line 1, then line 2, and so on round the wheel; the link needs 24 people.

Turn — a shift time, such as 06:20-13:45. An **early** starts before 09:00; a **late** starts after.

Cover (spare) week — a week with no fixed times. You work four duties of the seven days, placed by the roster clerk to cover leave and sickness.

The contract — 35 hours a week, averaged over the 24 lines; a cover week counts as a full contracted week.

Line — one week of the pattern, Sunday to Saturday. You move down a line each week, and line 24 goes back to line 1.

Rest day (RD) — a day off inside the contracted week. Two together make a proper break; one on its own does not count as a 48-hour break.

Sunday — not part of the contract. A Sunday duty is rest-day working, paid on top of the week, which is why the hours are shown Mon-Sat and all week.

One-turn week — the same clock time all week (shown bold in the Turns column); the easiest kind of week to live around.

One cell, sixteen line numbers

What the added Sunday buys, what the reorder buys, and the weekday column neither of them touches

Kept — the shape of a week

MEASURE	TODAY (20)	PROPOSED (24)
Working weeks that are one turn (one clock time Mon–Fri)	7 of 16	2 of 20
Weeks mixing early and late turns	7	8
Rest-day breaks that are two days or more	13 of 17	15 of 27
Single rest days between duties (not a 48h break)	4	12
Days worked in a week (lines × days)	1×3, 3×4, 11×5, 1×6	1×3, 4×4, 12×5, 3×6
Cover (spare) weeks (whole weeks)	lines 1, 7, 12, 17	lines 1, 7, 12, 17
Distinct shift times (all but 13 on today's roster)	18	24
Mon–Sat hours, average week (the contract)	35h 00m	35h 00m
Sunday duties (not contracted; half the working lines)	8 of 16 lines	10 of 20 lines

Changed — the December 2026 headcount

PEOPLE ON DUTY	TODAY	PROPOSED	WHY
Monday to Friday	11–12	13–16	the contract: 20 working lines × 35h has to be worked somewhere
Saturday	10	12	owner's figure for December 2026, leaning late for events
Sunday	8	10	owner's figure for December 2026
Opening at 06:20 (07:15 Sunday) wk · Sat · Sun	4 · 4 · 3	4 · 5 · 4	four to open, every day — not met: Saturday 5
Through to the close (23:55; 23:25 Sunday) wk · Sat · Sun	2 · 2 · 3	3–4 · 3 · 5	three to close, four on a Saturday — not met: Fri 4, Saturday 3, Sunday 5
Still on duty at 22:00 wk · Sat · Sun	4 · 2 · 3	4–6 · 5 · 5	five from 22:00, Monday to Saturday — not met: Wed 4, Fri 6

The duty table, in today's times

TURN	WK	SAT	SUN	WK	SAT	SUN
06:20–13:35 7h 15m · Early	1			1		
06:20–13:45 7h 25m · Early	1			3 +2		
06:20–14:00 7h 40m · Early		4			–4	
06:20–14:15 7h 55m · Early				1 +1		
06:20–14:20 8h 00m · Early	2			1 +1		
06:20–14:30 8h 10m · Early			2	2 +2		
06:20–14:50 8h 30m · Early		3		1 –2		
07:00–15:30 8h 30m · Early				2 +2		
07:15–15:15 8h 00m · Early						1 +1
07:15–15:45 8h 30m · Early			3			3
08:00–16:20 8h 30m · Early	–2			–2		
08:30–16:30 8h 00m · Early			1	1 +1		–1
08:30–17:00 8h 30m · Early				1 +1		
10:30–19:00 8h 30m · Early						1 +1
11:00–18:30 7h 30m · Late					1 +1	
11:00–19:30 8h 30m · Late	4			–4		
12:00–20:00 8h 00m · Late		4			–4	
12:00–20:30 8h 30m · Late				1 +1		
13:00–21:00 8h 00m · Late			1	1 +1		–1
13:30–21:00 7h 30m · Late		4			–4	
13:30–21:30 8h 00m · Late					1 +1	
13:30–22:00 8h 30m · Late	1			4 +3		
14:00–22:30 8h 30m · Late	2			2		
14:30–22:00 7h 30m · Late		–2			–2	
14:30–22:30 8h 00m · Late					2 +2	
14:30–23:25 8h 55m · Late			3			3
14:45–23:55 9h 10m · Late		–2			–2	
14:55–23:25 8h 30m · Late						1 +1
15:15–23:55 8h 40m · Late	2			–2	3 +3	
15:25–23:25 8h 00m · Late						1 +1
16:25–23:55 7h 30m · Late				4 +4		

Busiest weekday · Saturday · Sunday. The first three columns are what the 20-line link does now; the last three are the proposal. Green cells grew, amber shrank, a struck-through row is a time the proposal does not use.

Why the duties land where they do: Monday to Saturday still totals exactly 42,000 minutes — 20 × 35h to the minute — with the parent's daily split untouched: Mon 6,225 · Tue 7,205 · Wed 7,205 · Thu 7,780 · Fri 7,690 · Sat 5,895. **Sunday rises from 4,635 to 5,115 minutes**, ten duties where there were nine, and it sits on top of the contract as rest-day working, as every Sunday here does. The reorder moved no duty between days — each day's multiset of turns is exactly the parent's plus the one Sunday cell, asserted by the search rather than argued — so the coverage rows opposite differ from the parent's on Sunday alone.

People on duty, hour by hour

Cover today and proposed against the measured December 2026 timetable (arrivals and departures, weighted by train length). A darker orange hour carries more of the day's traffic.

How to read it: each blue cell is the number of people on duty in that hour (darker blue = more people). The orange row is the timetable — how much of the day's trains that hour carries (darker orange = busier). Read the columns top to bottom: cover should be thickest where the orange is darkest. The **fit** figure at the right end of each row scores that match for the whole day.

Monday to Friday window 06:20–23:55 · one row per distinct weekday

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY MON · WED		4	4	6	6	6	7	7	7	7	7	7	5	5	5	4	4	4	2	55.8
TODAY TUE · THU · FRI		4	4	6	6	6	7	7	8	8	8	8	6	6	6	5	5	4	2	50
PROPOSED MON		4	5	6	6	6	6	7	8	7	6	8	7	7	7	7	6	5	3	28.4
PROPOSED TUE		4	6	8	8	8	8	9	10	9	8	9	7	7	7	7	6	5	3	43.6
PROPOSED WED		4	5	7	7	7	7	8	11	9	8	10	8	8	8	8	7	4	3	42.3
PROPOSED THU		4	6	6	6	6	6	7	11	11	9	10	10	10	10	10	9	5	3	48.8
PROPOSED FRI		4	6	7	7	7	7	8	10	10	8	10	9	9	9	9	7	6	4	31
DEC 2026 TRAFFIC	5	69	107	127	122	95	85	75	84	79	87	105	140	138	107	90	73	75	66	

Saturday window 06:20–23:55

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY		4	4	4	4	4	4	5	6	9	6	6	6	6	6	5	4	2	2	45.7
PROPOSED		5	5	5	5	5	6	6	7	9	7	7	7	7	6	6	6	5	3	20.5
DEC 2026 TRAFFIC	6	24	58	75	81	92	88	77	64	68	72	84	82	67	64	79	78	62	41	

Sunday window 07:15–23:25

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY			3	4	4	4	4	4	5	8	8	5	4	4	4	4	3	3	3	65.9
PROPOSED			4	4	4	5	5	5	5	9	10	6	6	6	5	5	5	5	5	35.3
DEC 2026 TRAFFIC			15	45	65	71	74	71	61	55	64	69	78	73	75	52	71	65	50	

Spare cover is not in these figures — a cover week carries no times, so the rows are a floor. **Fit** is one number for how closely the people on duty follow the trains through the day: 0 would be a perfect match, lower is better, and today's link scores what it scores. (For the record: the squared gap between each hour's share of the day's traffic and its share of the cover, inside the day's window, measured on duty minutes — so an hour a duty only touches counts in proportion; the cells above count heads.) It is the measure the searched tables were chosen on, and the fit of the average weekday is the *Wk fit* on page 8. Sunday: 5 December 2026 timetable movements fall after the 23:25 finish — the standing question on whether Sunday's window moves; the window is stored per design, so this proposal can be rebuilt to either answer.

Reading it. The proposal raises the whole day rather than reshaping it, which is what the December 2026 timetable brief asked for — the station rosters posts (ticket office, gateline, passenger assist) plus break cover, and a post needs someone on it whether the hour carries 23 trains or 5. The weekday's two peaks are a duty length apart, so cover stays fairly flat between them by arithmetic, not neglect.

The checks sheet

Hard limits first — met or not. Then the ORR good-practice fatigue factors, which report what is present and never pass or fail a design.

Hard limits — a design either meets these or cannot be run

- ✓ **More than 13 consecutive days worked** — longest possible run **6** days (today: 9)
Longest possible run is 6 days, within the 13 configured here from Chiltern practice (origin: the legacy Hidden standard). A spare week is 4 duties of 7 and the roster clerk places them, so this is the worst case the link allows. Worked as a BLOCK of four rather than split day-on-day-off, the answer is the same — **6** days — so this row does not depend on how a cover week is placed.
Basis: Chiltern roster policy, citation outstanding — legacy Hidden 13-in-14 standard. Configured from Chiltern practice; the policy citation is outstanding, so this is stated as the app states it.
- ✓ **At least 12 hours between duties** — **0** rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today: 0)
Tightest anywhere: **12h 30m** — **line 13 Fri 14:00-22:30 into Sat 11:00-18:30** (today's tightest: 12h 30m). The generator refuses a design it cannot repair to this; this design was checked against it, not generated.
- ✓ **The contracted week, exactly** — **35h 00m** average Mon-Sat over the 24 lines, cover weeks counted as contracted weeks
700h of duty a week across 20 working lines. Sundays (85.25h) sit on top as RDW, as they do today. Individual weeks range 24h 25m–47h 10m; only the average is the contract.

December 2026 timetable design figures — the staffing shape agreed for the new timetable

RULE	PROPOSAL	
Four on at the open, every day	× 4 weekday · 5 Saturday · 4 Sunday	
Three through to the close; four on a Saturday	× 3–4 weekday · 3 Saturday · 5 Sunday	
Five still on duty at 22:00	× 4–6 weekday · 5 Saturday	a 22:00 finish is not "on at 22:00"
Fourteen on a Saturday, ten on a Sunday	× 12 · 10	
Cover weeks, evenly spread	× 4 weeks at lines 1, 7, 12, 17 — gaps 8, 6, 5, 5	the December 2026 staffing shape asks for four, evenly spread — the count is right; the spread is not
About 4.2 days a week worked, Mon-Sat	✓ 4.35 over the 20 working lines	
No :05 or :10 times except the open and close	✓ none	
Late turns slightly shorter than most earlies	× not met — lates 7h30–8h40, earlies 7h15–8h30	the longest late is longer than the longest early

Five things to settle before it is frozen

Sunday now closes with five, where the shape asks three. The parent already closed with four; the added turn finishes at 23:25 because 22:00 was Sunday's deepest remaining shortfall and the curve asked for exactly that. A 15:20–23:20 turn keeps the closers at four and costs 0.6 of fit — if the rule wins, that is the change. **Sixteen line numbers move.** Every week is intact, but a person's line number is what they know, and 16 of the 20 working weeks change theirs. The *Four swaps only* row opposite is the small version and it shows what small costs: run 8, three weekends, and a 55-hour window eighteen minutes from breaching. **The week that gains the Sunday.** The parent's week 24 — now line 5 — works Sunday 15:25–23:25, Monday off, then three earlies from Tuesday: a late into an early across one rest day. Legal by a wide margin (30h55) and a five-day week where it was four; it is the one week here whose shape a member would notice changing. **FF15 sits exactly on its threshold.** The longest run of early shifts is 4, and the factor reads "more than 4" — clear, not comfortably. **Everything the parent did not answer, it still does not.** Saturday opens with five and closes with three against four and four; Saturday carries 12 against 14; 4.35 days a week against 4.2; the longest late is 8h40 against a longest early of 8h30; Thursday is still the worst-fitting day at 48.8; two factors — a block of early starts with no rest after it, and start times that move more than two hours inside a block — remain present, and the *Full re-search* row shows what clearing one of them costs.

ORR fatigue factors

Good practice guidelines — Fatigue Factors, p3 (December 2021). Δ present — the pattern is in this design and worth a look, not a breach · $\checkmark$ clear — it is not · $\bullet$ standing — true of the station itself, not of any design · — does not apply here. Nathan assesses against this list.

The Office of Rail and Road lists 25 roster patterns that tend to tire people — long runs of earlys, short gaps between duties, start times that jump about. For each one this table asks whether the pattern is in the rotation, today and proposed, and how big it is. The list is guidance: a factor present is a question for the room, never a pass or a fail, and a design showing nothing is not thereby approved.

CODE	FACTOR	TODAY'S LINK	PROPOSED
FF1	Night shift covering 00:00–05:00 Time of day	--	--
FF2	Early shift starting 05:00–07:00 Time of day	$\bullet$ 24	$\bullet$ 25
FF3	Very early shift starting before 05:00 Time of day	$\checkmark$ 0	$\checkmark$ 0
FF4	Very early shift (before 05:00) over 8h Duty length	--	--
FF5	Day shift over 12h Duty length	$\checkmark$ 0	$\checkmark$ 0
FF7	Early shift over 10h Duty length	$\checkmark$ 0	$\checkmark$ 0
FF8b	Fewer than 2 days rest after a block of early starts Recovery time	Δ 3	Δ 3
FF11	More than 13 consecutive shifts without a 48h break Recovery time	$\checkmark$ 12 $\checkmark$ 9 as rostered	$\checkmark$ 12 $\checkmark$ 12 as rostered
FF10	More than 4 consecutive 12h day shifts Cumulative	$\checkmark$ 0	$\checkmark$ 0
FF15	More than 4 consecutive early shifts in a rotating pattern Cumulative	Δ 6	$\checkmark$ 4
MRSF	More than 12 consecutive day shifts Cumulative	$\checkmark$ 9	$\checkmark$ 6
MRSF	More than 55 hours worked in any 7-day period Cumulative	Δ 60.8	$\checkmark$ 50.9
MRSF	More than 7 consecutive 8h shifts (definition to confirm) Cumulative	Δ 9	$\checkmark$ 6
MRSF	More than 6 consecutive night or early shifts in a permanent pattern Cumulative	--	--
FF17	Backward rotating pattern (definition to confirm) Circadian	$\checkmark$ 10 backward / 12 forward	$\checkmark$ 13 backward / 20 forward
FF18	Rotating pattern of about a week (definition to confirm) Circadian	$\bullet$ 20-line rotation · typically 4h 0m a week	$\bullet$ 24-line rotation · typically 3h 10m a week
FF19	Successive start times varying by more than 2 hours (definition to confirm) Circadian	Δ 6	Δ 11
FF6	Night shift over 10h Night shifts	--	--
FF8	Less than 2 days rest after consecutive nights Night shifts	--	--
FF9	Less than 14h rest in any 24h for night shifts Night shifts	--	--
FF12	Only one day rest after night shifts Night shifts	--	--
FF14	More than 4 consecutive nights in a rotating pattern Night shifts	--	--
FF16	More than 3 consecutive night shifts over 8h Night shifts	--	--
FF20	First night shift Night shifts	--	--
FF13	Less than 12h rest in any 24h Intervals between duties	$\checkmark$ 0	$\checkmark$ 0

Present: 2 (today 5) · standing: 2 · FF2 fires on every 06:20 duty, so it is a property of the station's opening time rather than of any design. **Two readings of a cover week, on the one row they move.** A cover week is worked four days of seven and the link does not say which four, so a run figure is a range. The headline number is the ceiling — the four split day-on-day-off, which supplies no 48-hour break and joins the blocks either side. It is reachable: of the 35 ways to place four duties in seven days, the 10 that leave no two rest days together produce exactly it. Worked as a BLOCK, which is what a cover week looks like on the roster, the three rest days fall together and the week always supplies a break — that is the *as rostered* figure beneath it. Checked on every row: FF11 is the only one where the two differ. Which reading applies is a question for the roster office, not for this sheet. This sheet is an aid to a conversation, not a fatigue risk assessment.

METHOD

One cell, then a reorder

Why the Sunday shift could not simply be written in, what the reorder is, and the two other answers that were measured

In plain terms. Nobody searched for this design. It was typed in from the table exactly as supplied and the app's own rule modules measured it; nothing below was chosen by a computer. Everything on these pages can be re-checked by pasting page 9 into the Links workspace.

Where this design came from

1 • The cell. The ask was a tenth Sunday duty that fits December 2026 better. Every placement was measured — every line with a Sunday rest day, every start on the five-minute grid, four lengths, 1,771 placements clearing a 12h30 rest — and the demand curve is unambiguous: an evening turn, **15:25–23:25**, takes Sunday's misfit from 44.9 to **35.3**. But on the parent's line order that turn trips a factor *wherever* it goes: a worked Sunday joins the week before to the week after, so either FF11 (more than 13 shifts without a 48-hour break) or the 55-hour week appears. Only two lines could take any Sunday duty without a new factor, and both of them made the fit *worse*.

2 • The reorder. So the wheel was reordered — **whole weeks only** (optimise.mjs, LINES_ONLY=1), which is the smallest edit that can move a fatigue factor, because the factors are properties of which week follows which. Every week pattern is intact; 16 of the 20 working weeks change line number and 7 lines stay where they were; the week that gains the Sunday cell is the parent's week 24, now at

line 5. Result: factors present **3 → 2**, longest run **9 → 6**, full weekends off 4 → 5, the worst seven-day window 51.6h → 50.9h, tightest rest 12h30. Coverage, headcounts and the contract are invariant by construction, and the script asserts all three. It reproduces exactly from the committed start grid with the reorder search at seed 7 — the command is in the proposals README, beside the two other answers below.

3 • The two other answers, both in the table opposite. *Four swaps only* is the smallest reorder that reaches two factors: 17 lines untouched, but the run stays at 8, weekends fall to 3 and the 55-hour window sits at 54.7 — eighteen minutes from the line. *Full re-search* lets duties move between weeks as well: one factor, six weekends, 13 of 20 weeks a single turn — and 19 of the 20 working weeks rewritten. This proposal is the middle of those three; which end the room prefers is the room's call, not the search's. Not a recommendation.

Alternatives measured

DESIGN	RUN	FF PRESENT	WKENDS	ONE- TURN	STEP	WK FIT	SCORE
TN-24-R7 · 84b60df9 — Clean Final Ten (this proposal)	6	2	5	2/20	3h 10m	34.4	—
Same Turns · d15e1b74 — the searched proposal	6	1	6	18/20	2h 33m	46.3	—
By the Book · 0f14abce — the searched proposal	6	0	6	12/20	2h 52m	30.8	—
Tuned — the parent · ae1a15bd	9	3	4	2/20	3h 59m	34.4	—
Four swaps only · 3f925efe	8	2	3	2/20	4h 23m	34.4	—
Full re-search · f36f3aa3	6	1	6	13/20	2h 36m	34.4	—
Workspace default · 29ec9810 (Dec 2026 table, generated)	7	5	5	0/20	4h 16m	30.8	—
Today's 20-line link (for scale)	9	5	4	7/16	4h 0m	44.7	—

How to read this table: nothing here was picked by a search — this design was supplied or derived, and the rows are the comparisons it is being judged against. *Wk fit* is how evenly the WEEKDAY cover follows the December 2026 timetable traffic curve (lower is better; today's link scores what it scores), so it says nothing about Saturday or Sunday. **Score** is the search's own feel objective and is blank here, because no search produced this design — a blank is not a bad score. The workspace default is the app's own December 2026 duty table, generated and reordered with every switch on.

LOAD IT

The rotation, ready to paste

Line number, then Sunday to Saturday. SP is a cover week. Paste the whole block into Links → Import.

Links → Import, then paste the whole block below. Line number, then Sunday to Saturday; SP is a cover week. The workspace re-runs every check on these pages from the pasted cells, so nothing here has to be taken on trust — and the two designers can edit it there like any other design.

The same rotation is also supplied beside this PDF as Clean-Final-Ten-TN-24-R7-import.txt (tab-separated, pastes directly) and Clean-Final-Ten-TN-24-R7.json (the app's own format).

1	SP	SP	SP	SP	SP	SP	SP
2	07:15-15:15	06:20-13:45	RD	RD	13:30-22:00	16:25-23:55	13:30-21:30
3	14:55-23:25	13:30-22:00	RD	13:30-22:00	13:30-22:00	13:00-21:00	14:30-22:30
4	RD	16:25-23:55	16:25-23:55	RD	RD	07:00-15:30	06:20-14:30
5	15:25-23:25	RD	06:20-13:45	06:20-13:45	07:00-15:30	RD	06:20-14:15
6	14:30-23:25	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55	RD	RD
7	SP	SP	SP	SP	SP	SP	SP
8	14:30-23:25	RD	06:20-13:45	06:20-13:45	06:20-13:45	06:20-13:45	RD
9	RD	12:00-20:30	12:00-20:30	12:00-20:30	12:00-20:30	12:00-20:30	RD
10	RD	08:30-17:00	RD	16:25-23:55	16:25-23:55	16:25-23:55	15:15-23:55
11	RD	14:00-22:30	13:30-22:00	13:30-22:00	13:30-22:00	RD	RD
12	SP	SP	SP	SP	SP	SP	SP
13	07:15-15:45	06:20-13:45	RD	RD	13:30-22:00	14:00-22:30	11:00-18:30
14	10:30-19:00	16:25-23:55	RD	RD	06:20-14:20	06:20-14:20	06:20-14:30
15	07:15-15:45	RD	06:20-13:35	06:20-13:35	06:20-13:35	07:00-15:30	RD
16	RD	06:20-13:45	08:30-17:00	08:30-17:00	RD	RD	RD
17	SP	SP	SP	SP	SP	SP	SP
18	07:15-15:45	RD	08:30-16:30	06:20-14:20	06:20-14:20	06:20-14:20	06:20-14:20
19	RD	RD	14:00-22:30	13:30-22:00	14:00-22:30	14:00-22:30	14:30-22:30
20	14:30-23:25	RD	14:00-22:30	14:00-22:30	14:00-22:30	16:25-23:55	RD
21	RD	14:00-22:30	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55	15:15-23:55
22	RD	06:20-14:20	07:00-15:30	07:00-15:30	07:00-15:30	08:30-17:00	RD
23	RD	07:00-15:30	07:00-15:30	RD	RD	06:20-13:35	06:20-14:50
24	RD	RD	06:20-14:20	08:30-16:30	RD	13:30-22:00	15:15-23:55

Prepared 22 September 2026 · re-rendered 24 September 2026. The code **84b60df9** in every footer is the design's fingerprint — eight characters computed from these 168 cells and nothing else. If the workspace shows the same code after you paste, you have exactly this design; if one cell differs, the code changes. Every figure in this document was computed by the Marylebone Roster app's Links modules (v24.22) from exactly these cells; import them and the workspace's Design checks, hard limits, fatigue factors and coverage cards will restate them.